

# 1 Principles of Monte Carlo

## 1.1 Introduction Monte Carlo integration

Consider estimating  $\alpha = \int_0^1 f(x) dx$ . If  $U \sim \text{Unif}(0, 1)$ , then  $\alpha = \mathbb{E}[f(U)]$ . Given independent samples  $U_1, \dots, U_n \sim \text{Unif}(0, 1)$ , the Monte Carlo estimator is

$$\hat{\alpha}_n = \frac{1}{n} \sum_{i=1}^n f(U_i).$$

By the strong law of large numbers,  $\hat{\alpha}_n \rightarrow \alpha$  a.s. as  $n \rightarrow \infty$ . By the central limit theorem,

$$\hat{\alpha}_n - \alpha \approx N(0, \sigma_f^2/n),$$

where  $\sigma_f^2 = \text{var}(f(U))$ . Thus the standard error is  $\sigma_f/\sqrt{n}$ . Halving the standard error requires increasing the number of samples by a factor of 4.

### Dimensionality

For one-dimensional integration, Monte Carlo is usually not competitive with deterministic methods such as the trapezoidal rule, whose error is  $O(n^{-2})$ . The advantage of Monte Carlo is that its convergence rate is not directly sensitive to the dimension of the problem, while many deterministic integration methods suffer from the curse of dimensionality.

### 1.2 First Example: European Call Option

#### Risk-neutral valuation

Let  $S(t)$  be the stock price at time  $t$ . A European call option with strike  $K$  and maturity  $T$  has payoff  $(S(T) - K)^+ = \max\{0, S(T) - K\}$ . Under the risk-neutral measure  $\mathbb{Q}$ , the time-0 price is

$$V_0 = \mathbb{E}_0^{\mathbb{Q}}[e^{-rT} (S(T) - K)^+],$$

where  $r$  is the continuously compounded risk-free rate.

#### Stock-price dynamics

Under  $\mathbb{Q}$ , geometric Brownian motion satisfies  $\frac{dS(t)}{S(t)} = r dt + \sigma dW(t)$ , where  $W(t)$  is a standard Brownian motion. The exact solution at time  $T$  is

$$S(T) = S(0) \exp\left((r - \tfrac{1}{2}\sigma^2)T + \sigma W(T)\right).$$

Since  $W(T) = \sqrt{T}Z$  with  $Z \sim N(0, 1)$ ,

$$S(T) = S(0) \exp\left((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z\right).$$

#### Simulation algorithm

For  $i = 1, \dots, n$ , generate  $Z_i \sim N(0, 1)$ , then set

$$S_i(T) = S(0) \exp\left((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i\right),$$

$$C_i = e^{-rT} (S_i(T) - K)^+.$$

The Monte Carlo estimator is  $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$ . It is unbiased,  $\mathbb{E}[\hat{C}_n] = C$ , and strongly consistent,  $\hat{C}_n \rightarrow C$  with probability 1.

#### Confidence interval

The sample standard deviation of the simulated discounted payoffs is

$$s_C = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (C_i - \hat{C}_n)^2}.$$

An asymptotic  $1 - \delta$  confidence interval is

$$\hat{C}_n \pm z_{\delta/2} \frac{s_C}{\sqrt{n}}.$$

Since  $s_C$  is estimated, one may replace the normal quantile with the corresponding  $t$ -quantile with  $n - 1$  degrees of freedom.

#### Black-Scholes benchmark

The Black-Scholes value of a European call is

$$BS(S, \sigma, T, r, K) = S\Phi(d_1) - e^{-rT} K\Phi(d_2),$$

where

$$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}},$$

$$d_2 = \frac{\log(S/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

### 1.3 Simulation Layers

#### Layered construction

Monte Carlo derivative pricing is built in layers:

$$U_1 \rightarrow Z_1 \rightarrow S_1(T) \text{ or } \{S_i(t_j)\}_{j=1}^m \rightarrow C_i \rightarrow \hat{C}_n, s_C.$$

The lower layers transform simple random variables into asset prices or paths. The upper layers transform paths into discounted payoffs and then into statistical summaries.

### 1.4 Second Example: Asian Option

#### Path dependence

An Asian option is path dependent. A discrete-average Asian call has payoff  $(\bar{S} - K)^+$ , where

$$\bar{S} = \tfrac{1}{m} \sum_{i=1}^m S(t_i),$$

and  $0 = t_0 < t_1 < \dots < t_m = T$ . Its time-0 value is

$$\mathbb{E}_0^{\mathbb{Q}}[e^{-rT} (\bar{S} - K)^+].$$

To estimate this value, simulate the path values  $S(t_1), \dots, S(t_m)$ , compute the average  $\bar{S}$ , discount the payoff, and average over independent replications.

### 1.5 Efficiency of Simulation Estimators

#### Central limit theorem

Suppose  $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$ , where  $C_i$  are i.i.d.,  $\mathbb{E}_0^{\mathbb{Q}}[C_i] = C$ , and  $\text{var}(C_i) = \sigma_C^2 < \infty$ . Then

$$\frac{\hat{C}_n - C}{\sigma_C/\sqrt{n}} \Rightarrow N(0, 1),$$

or equivalently

$$\sqrt{n}(\hat{C}_n - C) \Rightarrow N(0, \sigma_C^2).$$

Hence  $\hat{C}_n - C \approx N(0, \sigma_C^2/n)$ . For two unbiased estimators of the same quantity, the estimator with lower variance is preferred, other things being equal.

### 1.6 Bias

#### General principle

Variance reduction and faster computation are useful only if the estimator converges to the correct value. Some estimators are biased for every finite  $n$ , but become asymptotically unbiased as computational effort increases.

#### Ratio estimator

For i.i.d. samples  $(X_i, Y_i)$ , the estimator  $\bar{X}/\bar{Y}$  for  $\mathbb{E}[X]/\mathbb{E}[Y]$  is generally biased because

$$\mathbb{E}\left[\frac{\bar{X}}{\bar{Y}}\right] \neq \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}.$$

However,

$$\frac{\bar{Y}}{\bar{X}} \rightarrow \frac{\mathbb{E}[Y]}{\mathbb{E}[X]}$$

with probability 1, and the normalized error is asymptotically normal. The bias becomes negligible as  $n \rightarrow \infty$ .

#### Model discretization error

If the exact geometric Brownian motion transition is replaced by the Euler recursion

$$S((j+1)h) = S(j)h + rS(j)h + \sigma S(j)h\sqrt{h}Z_{j+1},$$

then the simulated path distribution is not exactly the same as that implied by the continuous-time model. Option values estimated from this approximate dynamics are biased.

This model discretization bias can be reduced or eliminated by: (i) using the exact transition when available; (ii) decreasing the time step  $h$ , at the cost of more transitions per path.

#### Payoff discretization error

For a continuously averaged Asian option,

$$\bar{S} = \tfrac{1}{T} \int_0^T S(u) du.$$

Even exact simulation at a finite set of dates cannot compute the continuous average exactly. The payoff must be approximated, which creates payoff discretization error. This bias can be made small by using a larger simulation time step, but computational cost increases.

Model discretization error comes from approximating the underlying dynamics. Payoff discretization error comes from approximating the payoff functional.

### 1.7 Mean Square Error

#### Bias-variance decomposition

For an estimator  $\hat{\alpha}$  of  $\alpha$ , the mean square error is

$$\text{MSE}(\hat{\alpha}) = \mathbb{E}\left[(\hat{\alpha} - \alpha)^2\right].$$

It decomposes as

$$\text{MSE}(\hat{\alpha}) = (\mathbb{E}[\hat{\alpha}] - \alpha)^2 + \mathbb{E}\left[(\hat{\alpha} - \mathbb{E}[\hat{\alpha}])^2\right],$$

or

$$\text{MSE}(\hat{\alpha}) = \text{Bias}^2(\hat{\alpha}) + \text{Variance}(\hat{\alpha}).$$

Thus MSE balances bias and variance.

# 2 Generating Random Numbers and Random Variables

### 2.1 Random Number Generation

#### Uniform inputs

Monte Carlo simulation is driven by apparently random numbers. Ideally we want  $U_1, U_2, \dots$  such that each  $U_i \sim \text{Unif}[0, 1]$  and the  $U_i$  are mutually independent. Independence is crucial:  $U_i$  should not be predictable from  $U_1, \dots, U_{i-1}$ .

A pseudorandom number generator produces a finite deterministic sequence  $u_1, \dots, u_M \in [0, 1]$ . A good generator makes short segments hard to distinguish from independent uniforms. Desirable properties: long period, reproducibility, speed, portability, and statistical randomness.

#### Linear congruential generator

A linear congruential generator is

$$x_{i+1} = ax_i \bmod m, \quad u_{i+1} = x_{i+1}/m.$$

Here  $a$  is the multiplier,  $m$  the modulus, and  $x_0 \in \{1, \dots, m - 1\}$  the seed. The modulo operation is  $y \bmod m = y - \lfloor y/m \rfloor m$ .

Once a value repeats, the whole sequence repeats. The generator has full period if it produces all  $m - 1$  nonzero values before repeating. Large  $m$  alone is not enough: poor choices of  $a, m$  can produce short cycles.

### 2.2 General Sampling Methods

#### Uniform distribution

Assume independent uniforms satisfying  $P(U_i \leq u) = 0$  if  $u < 0$ ,  $P(U_i \leq u) = u$  if  $0 \leq u \leq 1$ , and  $P(U_i \leq u) = 1$  if  $u > 1$ . Sampling methods transform these uniforms into variables with desired distributions.

#### 2.3 Inverse Transform Method

##### General method

To sample  $X$  with CDF  $F$ , set

$$X = F^{-1}(U), \quad U \sim \text{Unif}[0, 1].$$

If  $F$  is not strictly increasing, use the generalized inverse

$$F^{-1}(u) = \inf\{x : F(x) \geq u\}.$$

This chooses the smallest compatible  $x$ . Validity follows from

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x).$$

Flat sections of  $F$  have zero probability: if  $F$  is constant on  $[a, b]$ , then  $P(a < X \leq b) = F(b) - F(a) = 0$ .

#### Examples

Exponential with mean  $\theta$ :  $F(x) = 1 - e^{-x/\theta}$  for  $x \geq 0$ , so

$$X = -\theta \log(1 - U), \text{ equivalently } X = -\theta \log U.$$

Rayleigh example:

$$F(x) = 1 - e^{-2x(x-b)}, \quad x \geq b,$$

so the inverse-transform sample is

$$X = \tfrac{b}{2} + \sqrt{\tfrac{b^2 - 2\log U}{2}}.$$

Discrete distribution: for outcomes  $c_1 < \dots < c_n$  with probabilities  $p_i$ , define  $q_0 = 0$  and

$$q_i = \sum_{j=1}^i p_j.$$

Generate  $U$ , find  $K$  with  $q_{K-1} < U \leq q_K$ , and set  $X = c_K$ . Binary search can implement the lookup.

#### Conditional sampling

To sample  $X$  conditional on  $a < X \leq b$ , with  $F(a) < F(b)$ , generate

$$V = F(a) + [F(b) - F(a)]U.$$

Then  $F^{-1}(V)$  has the conditional distribution because

$$P(F^{-1}(V) \leq x) = \frac{F(x) - F(a)}{F(b) - F(a)}.$$

### 2.4 Acceptance-Rejection Method

#### General method

To sample from a target density  $f$  on  $\mathcal{X}$ , choose a convenient proposal density  $g$  and a constant  $c$  such that

$$f(x) \leq c g(x), \quad x \in \mathcal{X}.$$

Generate  $X \sim g$  and  $U \sim \text{Unif}[0, 1]$ . Accept  $X$  if

$$U \leq \frac{f(X)}{c g(X)}.$$

otherwise reject and repeat. If  $Y$  is the accepted value, then  $P(Y \in A) = P(X \in A) \mid U \leq f(X)/(cg(X))$ .

The acceptance probability is

$$P(U \leq f(X)/(cg(X))) = \int_{\mathcal{X}} \frac{f(x)}{c g(x)} g(x) dx = 1/c.$$

Hence

$$P(Y \in A) = c \int_A \frac{f(x)}{c g(x)} g(x) dx = \int_A f(x) dx.$$

Thus  $Y$  has density  $f$ . The number of proposals until acceptance is geometric with mean  $c$ , so smaller  $c$  and tighter bounds are better.

#### Beta distribution

The beta density on  $[0, 1]$  is

$$f(x) = \frac{1}{B(\alpha_1, \alpha_2)} x^{\alpha_1-1} (1-x)^{\alpha_2-1}.$$

Here

$$B(\alpha_1, \alpha_2) = \int_0^1 x^{\alpha_1-1} (1-x)^{\alpha_2-1} dx = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2)},$$

and

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx, \quad \text{Re}(z) > 0.$$

For integer  $n > 0$ ,  $\Gamma(n) = (n-1)!$ . If  $\alpha_1, \alpha_2 \geq 1$  and at least one exceeds 1, the mode is  $\frac{\alpha_1-1}{\alpha_1+\alpha_2-2}$ .

Let  $c$  be the maximum value of  $f$  and use  $g(x) = 1$  on  $[0, 1]$ . Generate  $U_1, U_2$  until  $cU_2 \leq f(U_1)$ , then return  $U_1$ .

#### Normal from double exponential

For the double exponential density  $g(x) = e^{-|x|}/2$  and standard normal density  $f(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$ ,

the ratio satisfies

$$\frac{f(x)}{g(x)} = \sqrt{\tfrac{\pi}{2}} e^{-x^2/2+|x|} \leq \sqrt{\tfrac{\pi}{2}} \approx 1.3155 \approx c.$$

The rejection test may be written as

$$u > \exp\left(-\tfrac{1}{2}x^2 + |x| - \tfrac{1}{2}\right) = \exp\left(-\tfrac{1}{2}(|x| - 1)^2\right).$$

Using symmetry: generate  $U_1, U_2, U_3$ , set  $X = -\log U_1$ , reject if  $U_2 > \exp[-0.5(X - 1)^2]$ , then set  $X \leftarrow -X$  if  $U_3 \leq 0.5$ .

### 2.5 Normal Random Variables and Vectors

#### Multivariate normal

A  $d$ -dimensional normal is written  $N(\mu, \Sigma)$ , where  $\mu$  is a  $d$ -vector and  $\Sigma$  is a  $d \times d$  covariance matrix. A covariance matrix must be symmetric and positive semidefinite:

$$x^\top \Sigma x \geq 0, \quad x \in \mathbb{R}^d.$$

If  $\Sigma$  is positive definite, the density is

$$\phi_{\mu, \Sigma}(x) = \frac{1}{(2\pi\rho/2)^{d/2}|\Sigma|^{1/2}} \exp\left[-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right].$$

If  $X \sim N(\mu, \Sigma)$ , then  $X_i \sim N(\mu_i, \sigma_i^2)$  with  $\sigma_i^2 = \Sigma_{ii}$ , and

$$\text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \Sigma_{ij}.$$

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}, \text{ equivalently } \Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}.$$

If  $\Sigma$  is positive semidefinite but singular, define  $X = \mu + AZ$  with  $Z \sim N(0, I_d)$  and  $AA^\top = \Sigma$ . Then  $X$  has no full  $d$ -dimensional density when  $\text{rank}(\Sigma) < d$ .

#### Useful properties

Linear transformation: if  $X \sim N(\mu, \Sigma)$ , then

$$AX \sim N(A\mu, A\Sigma A^\top).$$

Conditioning formulas: for a partitioned normal vector with  $\Sigma_{[22]}$  full rank,

$$(X_{[1]} \mid X_{[2]} = x) \sim N(\mu_{[1]} + \Sigma_{[12]}\Sigma_{[22]}^{-1}(x - \mu_{[2]}), \Sigma_{[11]} - \Sigma_{[12]}\Sigma_{[22]}^{-1}\Sigma_{[21]}).$$

Moment generating function:

$$\mathbb{E}[\exp(\theta^\top X)] = \exp(\mu^\top \theta + \tfrac{1}{2}\theta^\top \Sigma \theta).$$

### 2.6 Generating Normals

#### Box-Muller method

The Box-Muller method generates a bivariate normal distribution. If  $Z \sim N(0, I_2)$ ,

then  $R = Z_1^2 + Z_2^2$  is exponential with mean 2:

$$P(R \leq x) = 1 - e^{-x/2}.$$

Given  $R, (Z_1, Z_2)$  is uniformly distributed on the circle of radius  $\sqrt{R}$ . Generate independent  $U_1, U_2$ , set

$$R = -2\log U_1, \quad V = 2\pi U_2,$$

and return

$$Z_1 = \sqrt{R} \cos V, \quad Z_2 = \sqrt{R} \sin V.$$

#### Multivariate normals

To sample  $X \sim N(\mu, \Sigma)$ , generate  $Z \sim N(0, I_d)$ , find  $A$  such that

$$AA^\top = \Sigma,$$

and set  $X = \mu + AZ$ . Then  $X \sim N(\mu, \Sigma)$ .

### 2.7 Cholesky Factorization

#### Positive definite case

Cholesky represents  $\Sigma = AA^\top$  with  $A$  lower triangular. For

$$\Sigma = \begin{pmatrix} \sigma_1^2 & & \\ \sigma_1 \sigma_2 \rho & \sigma_2^2 \rho^2 \\ \sigma_1 \sigma_2 \rho & \sigma_2^2 \rho^2 \end{pmatrix},$$

a Cholesky factor is

$$A = \begin{pmatrix} \sigma_1 & & 0 \\ \rho \sigma_2 & \sqrt{1-\rho^2} \sigma_2 \end{pmatrix}.$$

Thus

$$X_1 = \mu_1 + \sigma_1 Z_1,$$

$$X_2 = \mu_2 + \sigma_2 \rho Z_1 + \sigma_2 \sqrt{1-\rho^2} Z_2.$$

For general  $d$ , use

### Optimization: Mean-Semi-Variance Portfolio Selection

The data-driven mean-semi-variance portfolio selection can be formulated as the following quadratic program(no borrowing and no shorting.):

$$\begin{aligned} \min_{w_1, \dots, w_T} \quad & \frac{1}{T} \sum_{t=1}^T y_t^2 \\ \text{s.t.} \quad & y_t \geq w^\top (\bar{R} - R_t), \quad t = 1, \dots, T \\ & y_t \geq 0, \quad t = 1, \dots, T \\ & r_f + w^\top (R - r_f \mathbf{1}_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1, \dots, n \end{aligned}$$

### Optimization: Mean-Absolute Standard Deviation Portfolio Selection

The data-driven mean-absolute standard deviation portfolio selection can be formulated as the following linear program(no borrowing and no shorting):

$$\begin{aligned} \min_{w_1, \dots, w_T} \quad & \frac{1}{T} \sum_{t=1}^T y_t \\ \text{s.t.} \quad & y_t \geq w^\top (R_t - \bar{R}), \quad t = 1, \dots, T \\ & y_t \geq -w^\top (R_t - \bar{R}), \quad t = 1, \dots, T \\ & r_f + w^\top (R - r_f \mathbf{1}_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1, \dots, n \end{aligned}$$

### Optimization: Mean-CVaR Portfolio Selection

Suppose the initial wealth is  $x_0$ , so for portfolio  $w$  the P&L is  $Z_w = x_0 R_w$ . Suppose the target for the expected return rate is  $z$ . The data-driven mean- $\alpha$ -CVaR portfolio selection can be formulated as the following linear program:

$$\begin{aligned} \min_{w_1, \dots, w_T, b} \quad & \frac{1}{(1-\alpha)T} \sum_{t=1}^T y_t \\ \text{s.t.} \quad & y_t \geq -(r_f + w^\top (R_t - r_f \mathbf{1}_n))x_0 - b, \quad t = 1, \dots, T \\ & y_t \geq 0, \quad t = 1, \dots, T \\ & r_f + w^\top (\bar{R} - r_f \mathbf{1}_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1, \dots, n \end{aligned}$$

### 4.7 Expected Utility

Expected utility (EU) theory suggests that when making decisions under risk, people do not evaluate monetary payoffs directly. Instead, they attribute a subjective value (utility) to each monetary payoff.  $X$  is evaluated as the expectation of its utility  $\mathbb{E}[u(X)]$ . With a logarithmic utility  $u(x) = \log(x)$ , the expected utility of the St. Petersburg game is:

$$\mathbb{E}[u(X)] := \sum_{n=1}^{\infty} (1/2^n) \times \log(2^n) = \log 2 \times \sum_{n=1}^{\infty} n/2^n = \log(4)$$

This solves the paradox.

### Risk Aversion

Individuals are risk averse if any random payoff  $X$  is less preferred to its mean  $\mathbb{E}[X]$ . An individual with EU preferences is risk averse if and only if their utility function  $u$  is concave. Concavity is equivalent to diminishing sensitivity, meaning the marginal utility  $u'$  is decreasing.

### Certainty Equivalent

The certainty equivalent of a random payoff  $X$  is the deterministic payoff  $c$  that is indifferent to  $X$ :

$$u(c) = \mathbb{E}[u(X)]$$

Individual  $A$  is more risk averse than  $B$  if the certainty equivalent of  $X$  for  $A$  is less than or equal to that for  $B$ .

### Risk Aversion Coefficient

The Arrow-Pratt absolute risk-aversion coefficient is defined as:

$$-\frac{u''(x)}{u'(x)}$$

The larger the coefficient, the more concave the utility function and the more risk averse the individual.

### Equivalent Utility Functions

Preferences remain the same if we replace  $u(x)$  with  $au(x) + b$  where  $a > 0$ .

Thus,  $v(x) := au(x) + b$  is equivalent to  $u(x)$ .

### Examples of Utility Functions

Exponential utility ( $\gamma > 0$ ):

$$u(x) = 1 - e^{-\gamma x}, \quad x \in \mathbb{R}$$

Power utility ( $\gamma > 0, \gamma \neq 1$ ):

$$u(x) = x^{1-\gamma} / (1-\gamma), \quad x > 0$$

Logarithmic utility:

$$u(x) = \ln x, \quad x > 0$$

Quadratic utility ( $\gamma > 0$ ):

$$u(x) = x^2, \quad x \in \mathbb{R}$$

### 4.8 Elicit Utility Function

#### Certainty Equivalent Method

Select reference points  $A$  and  $B$ . Propose lottery  $(A, p; B, 1-p)$ . Ask for certainty equivalent  $c$ . Then:

$$u(c) = pu(A) + (1-p)u(B)$$

$u(A)$  and  $u(B)$  are set exogenously due to scaling freedom. Varying  $p$  maps the curve.

#### Probability Equivalent Method

Select reference points  $A$  and  $B$ . For a value  $x$ , ask for the probability  $p$  such that the individual accepts the lottery in place of  $x$ . Then:

$$u(x) = pu(A) + (1-p)u(B)$$

#### Tradeoff Method

Compare lotteries  $(X, p; r)$  and  $(x, p; R)$  with reference outcomes  $R > r$ . Vary  $x$  until indifference. Then compare  $(Y, p; r)$  and  $(y, p; R)$ . By EUT:

$$u(X) - u(x) = u(Y) - u(y)$$

This establishes an indifference relation. Starting from a base  $x_0$ , one can identify a sequence where  $u(x_{i+1}) = ju(x_i)$ .

### 4.9 Portfolio Selection with EU in Single Period

Consider a single period problem with initial wealth  $x_0$ , assets with returns  $R_i$ , and portfolio weights  $w_i$ . Portfolio return is  $R_w = \sum_{i=1}^n w_i R_i$ . Terminal wealth is  $x_0(1 + R_w)$ . The objective is:

$$\begin{aligned} \max_{w_1, \dots, w_n} \quad & \mathbb{E}[u(x_0(1 + R_w))] \\ \text{s.t.} \quad & \sum_{i=1}^n w_i = 1 \end{aligned}$$

### Relation with Mean-Variance Criterion

For quadratic utility  $u(x) = x - (\gamma/2)x^2$ , the expected utility depends on  $\mathbb{E}[R_w]$  and  $\text{Var}[R_w]$ . Thus, the optimal portfolio must be in the mean-variance set. Expected utility of terminal wealth:

$$\begin{aligned} & x_0 \left\{ \mathbb{E}[1 + R_w] - (\gamma x_0/2) \mathbb{E}[(1 + R_w)^2] \right\} \\ = & x_0 \left\{ 1 - (\gamma x_0/2) + (1 - \gamma x_0) \mathbb{E}[R_w] + (\gamma x_0/2) (\mathbb{E}[R_w])^2 - (\gamma x_0/2) \text{Var}[R_w] \right\} \end{aligned}$$

For exponential utility  $u(x) = -e^{-\gamma x}$  and normally distributed returns, the expected utility is:

$$-\exp\left(-\gamma x_0(1 + \mathbb{E}[R_w]) + \frac{\gamma^2 x_0^2}{2} \text{Var}[R_w]\right)$$

Thus, the optimal portfolio must be on the mean-variance efficient frontier.

### General Case

In general, EU and mean-variance criteria differ. Using the Lagrange dual method, the optimal portfolio must  $w$  satisfies:

$$\begin{aligned} \mathbb{E}[u'(x_0(1 + R_w))] \lambda_0 R_i] &= \lambda, \quad i = 1, \dots, n \\ \text{s.t.} \quad & \sum_{i=1}^n w_i = 1 \end{aligned}$$

## 5 CAPM and Arbitrage Pricing Theory

### 5.1 Market Equilibrium and CAPM model

**Market equilibrium:** at equilibrium price

All investors have achieved their goals. Market is clear, demand=supply.

**Capital asset pricing model (CAPM):** an equilibrium pricing model in framework of mean-variance analysis.

#### Assumptions

Each investor is a price-taker and mean-variance optimizer (although with different risk appetite). Everyone agrees on the probabilistic structure of assets (mean, covariance matrix of future asset prices). There is a unique risk-free rate  $r_f$  of borrowing and lending that is available to all. Market is frictionless: assets can be traded in any fraction and there is no transaction fee and other fees. The equilibrium price exists.

#### Market Portfolio

Suppose there are  $n$  risky assets in the market with number of shares  $N_i, i = 1, \dots, n$ . The current prices are  $P_i, i = 1, \dots, n$ .

Total capital:  $\sum_{i=1}^n N_i P_i$

Capitalization weight:  $\bar{w}_M = (\bar{w}_M, 1, \dots, \bar{w}_M, \bar{w}_M)$   
 $\bar{w}_M = \frac{\text{capital in asset } i}{\text{total capital}} = \frac{N_i P_i}{\sum_{i=1}^n N_i P_i}, i = 1, \dots, n$

Market portfolio: portfolio with capitalization weight.

#### Market Portfolio and Tangent Portfolio

Each investor holds his optimal portfolio which is a mixture of risk-free asset and mutual tangent portfolio. The total assets held by all investors must be the mixture of risk-free asset and tangent portfolio. Under equilibrium, total holding of assets equals total supply. **Thus tangent portfolio is market portfolio.**

Suppose there are  $J$  investors in the market. Investor  $j$  has initial wealth  $x_j$  and invests in the portfolio  $(w_{j,0}, (1 - w_{j,0})\bar{w}_P)$ ,  $j = 1, \dots, J$ . The number of shares of the  $i$ -th risky asset investor  $j$  holds is  $x_j(1 - w_{j,0})\bar{w}_{P,i}/P_i, i = 1, \dots, n$ . In equilibrium:

$$\left( \sum_{j=1}^J x_j(1 - w_{j,0}) \right) \bar{w}_{P,i}/P_i = N_i$$

Thus,  $\sum_{j=1}^J x_j(1 - w_{j,0}) = \sum_{i=1}^n N_i P_i$ , and for  $i = 1, \dots, n$ :

$$\bar{w}_{P,i} = \frac{N_i P_i}{\sum_{j=1}^J x_j(1 - w_{j,0})} = \frac{N_i P_i}{\sum_{i=1}^n N_i P_i} = \bar{w}_{M,i}, \quad i = 1, \dots, n$$

#### Capital Market Line

The efficient set is a single straight line starting from the risk-free point and passing through the market portfolio. This line is called capital market line. Mathematically, capital market line states for any efficient portfolio of assets, its return  $R$  satisfies:

$$\mathbb{E}[R] = r_f + \frac{\mathbb{E}[R_M] - r_f}{\sigma_M} \sigma$$

where  $\mathbb{E}[R]$  and  $\sigma$  are the mean and standard deviation of  $R$ , respectively, and  $\mathbb{E}[R_M]$  and  $\sigma_M$  are the mean and standard deviation of the return of the market portfolio.

A short proof: Efficient portfolio  $(w_0, (1 - w_0)\bar{w}_M)$ ; its expected return is:

$$\mathbb{E}[R] = w_0 r_f + (1 - w_0) \mathbb{E}[R_M] = r_f + (\mathbb{E}[R_M] - r_f)(1 - w_0)$$

and its deviation is:

$$\sigma = (1 - w_0) \sigma_M$$

Thus, we obtain the capital market line (assuming  $\tau_M \geq r_f$ ). Capital market line is the efficient frontier.

#### Derivation of Asset Pricing

Consider asset/portfolio  $i$  with return  $R_i$ , and denote its expectation and standard deviation as  $\mathbb{E}[R_i]$  and  $\sigma_i$ , respectively. Consider the mean-variance problem with  $z = \mathbb{E}[R_i]$ . The optimal portfolio is:

$$\begin{bmatrix} w_0 \\ (1 - w_0) \bar{w}_M \end{bmatrix}$$

$$\text{with } 1 - w_0 = \frac{\mathbb{E}[R_i] - r_f}{\mathbb{E}[R_M] - r_f}.$$

For any  $\epsilon \in \mathbb{R}$ , consider the portfolio which is mixture of  $(1 - \epsilon)$  of the optimal portfolio and of  $\epsilon$  of asset  $i$ . This new portfolio is feasible with expected return  $z$  and its standard deviation is:

$$\begin{aligned} & \text{Var}((1 - \epsilon)w_0 r_f + (1 - \epsilon)(1 - w_0)R_M + \epsilon R_i) \\ = & (1 - \epsilon)^2 (1 - w_0)^2 \text{Var}(R_M) + 2\epsilon(1 - \epsilon)(1 - w_0) \text{cov}(R_i, R_M) + \epsilon^2 \text{Var}(R_i) \\ = & f(\epsilon) \end{aligned}$$

0 is the minimum point of  $f(\cdot)$  because  $(w_0; (1 - w_0)\bar{w}_M)$  is in the minimum-variance set.

#### Capital Asset Pricing Model (CAPM)

First-order condition:

$$0 = f'(\epsilon) = -2(1 - w_0)^2 \text{Var}(R_M) + 2(1 - w_0) \text{cov}(R_i, R_M)$$

leads to:

$$1 - w_0 = \frac{\text{cov}(R_i, R_M)}{\text{Var}(R_M)}$$

Capital asset pricing model (CAPM):

$$\mathbb{E}[R_i] - r_f = \beta_i (\mathbb{E}[R_M] - r_f)$$

where

$$\begin{aligned} \beta_i &= \frac{\text{cov}(R_i, R_M)}{\text{Var}(R_M)} \\ \mathbb{E}[R] - r_f &= \beta (\mathbb{E}[R_M] - r_f) \end{aligned}$$

draws security market line on  $(\beta, \mathbb{E}[R])$  plane.

$R_i$  is referred to as the beta of the asset.  $\mathbb{E}[R_i] - r_f$  is termed as expected excess rate of return of asset  $i$ ;  $\mathbb{E}[R_M] - r_f$  is the expected excess rate of return of market portfolio.

CAPM says expected excess return of an asset is proportional to expected excess return of market portfolio. The proportion is its beta  $(\beta)$ .

$\beta > 0$ , i.e., positively correlated; positive expected excess return.

$\beta = 0$ , i.e., uncorrelated; zero expected excess return.

$\beta < 0$ , i.e., negatively correlated; negative expected excess return.

It is  $\beta$  rather than  $\sigma$  that contributes to expected excess return.

### Economic Interpretation of CAPM

Consider the portfolio consisting of market portfolio,  $\epsilon$  of asset  $i$ , and  $-\epsilon$  of risk-free asset, the last two components being a zero cost trading strategy with P&L  $\epsilon(R_i - r_f)$ . The variance of the portfolio is:

$$\text{Var}(r_f \epsilon (1 + R_w) + \epsilon \text{cov}(R_i, R_M) + \epsilon^2 \text{Var}(R_i))$$

When  $\epsilon \ll 1$ , second term dominates third term. If  $\text{cov}(R_i, R_M) = 0$ , asset  $i$  does not affect the variance, just like risk-free asset; thus it has zero risk premium. If  $\text{cov}(R_i, R_M) > 0$ , asset  $i$  increases the variance; thus it has positive risk premium. If  $\text{cov}(R_i, R_M) < 0$ , asset  $i$  reduces the variance; thus it has negative risk premium.

Key insight: risk is the variance of the portfolio return, not the variance of any single asset's return.

#### Decompose the Risk

For any asset  $i$  with (random) return  $R_i$ , define  $\epsilon_i$  via:

$$R_i - r_f = \beta_i (R_M - r_f) + \epsilon_i$$

By CAPM,  $\mathbb{E}[\epsilon_i] = 0$ . By the definition of  $\beta_i$ :

$$\begin{aligned} \text{cov}(R_M, \epsilon_i) &= \text{cov}(R_M, R_i - r_f - \beta_i (R_M - r_f)) \\ &= \text{cov}(R_M, R_i - r_f) - \text{cov}(R_M, \beta_i R_M - \beta_i r_f) \\ &= \text{cov}(R_M, R_i) - \beta_i \text{cov}(R_M, R_M) = 0 \end{aligned}$$

The risky return  $R_i$  can be decomposed into two parts: the part comoves with market return and the part uncorrelated with market return. As a result:

$$\sigma_i^2 = \text{cov}(\beta_i (R_M - r_f) + \epsilon_i, \beta_i (R_M - r_f) + \epsilon_i) = \beta_i^2 \sigma_M^2 + \text{Var}(\epsilon_i)$$

The first term,  $\beta_i^2 \sigma_M^2$  is called **systematic risk**. The second term,  $\text{Var}(\epsilon_i)$ , is called **nonsystematic, idiosyncratic, specific risk**. Only the systematic risk contributes to expected return according to CAPM. Any asset or portfolio on capital market line (efficient frontier) has only systematic risk. Any asset or portfolio below capital market line has nonsystematic risk.

#### CAPM as a Pricing Formula

Certainly equivalent pricing formula: The price  $P$  of an asset with payoff  $Q$  is:

$$P = \frac{1}{1+r_f} \frac{\mathbb{E}[Q] - \frac{\text{cov}(Q, R_M)}{\text{Var}(R_M)} (\mathbb{E}[R_M] - r_f)]{\mathbb{E}[R_M] - r_f}$$

Pricing form of CAPM: The price  $P$  of an asset with payoff  $Q$  (random variables) is:

$$P = \frac{\mathbb{E}[Q]}{1+r_f + \beta (\mathbb{E}[R_M] - r_f)}$$

First formula is used when the correlation of future payoff with market portfolio is known. Second formula is used when  $\beta$  is known a priori.

#### Example: The Oil Venture

Example 7.6 (The oil venture) Consider the possibility of investing in a share of a certain oil well that will produce a payoff that is random. The expected payoff is \$1,000 and the standard deviation of return is a relatively high 40%. The beta of the asset is  $\beta = 6$ . The risk-free rate is  $r_f = 10\%$ , and the expected return of the market portfolio is .17. What is the value of this share of the oil venture, based on CAPM? From the pricing formula, we have:

$$P = \frac{\mathbb{E}[Q]}{1+r_f + \beta (\mathbb{E}[R_M] - r_f)} = \frac{\$1,000}{1+0.10 + 6.0 \times (0.17 - 0.1)} = \$876$$

#### Example: ABC Fund Analysis

Example 7.4 (ABC fund analysis) The ABC mutual fund has a 10-year record of rates of return. We would like to evaluate this fund's performance in terms of mean-variance portfolio theory and the CAPM.

Step 1 Estimate  $\mu := \mathbb{E}[R]$ ,  $r_f, \mu_M := \mathbb{E}[R_M]$ , and  $\beta$  from the data:

$$\begin{aligned} \hat{\mu} &= \frac{1}{n} \sum_{t=1}^n R^t, \quad \hat{\mu}_M = \frac{1}{n} \sum_{t=1}^n R_M^t, \quad \hat{r}_f = \frac{1}{n} \sum_{t=1}^n r_f^t \\ \text{cov}(R^t, R_M^t) &= \frac{1}{n-1} \sum_{t=1}^n (R^t - \bar{R})(R_M^t - \bar{\mu}_M) \\ \text{Var}(R_M) &= \frac{1}{n-1} \sum_{t=1}^n (R_M^t - \bar{\mu}_M)^2 \end{aligned}$$

$$\begin{aligned} \hat{\beta} &= \frac{\text{cov}(R^t, R_M^t)}{\text{Var}(R_M^t)} \end{aligned}$$

Step 2 We write the formula:

$$\hat{\mu} - \hat{r}_f = J + \hat{\beta} (\hat{\mu}_M - \hat{r}_f)$$

If ABC fund is priced according to CAPM,  $J$  should be zero statistically.  $J$  is known as Jensen's index or alpha ( $\alpha$ ). For ABC fund,  $J > 0$ . If you are a CAPM believer, you may think ABC fund is underpriced and thus a good asset. If you are market efficiency believer, you may think CAPM is invalid.

Step 3 Find the estimate of the Sharpe ratio:

$$S := \frac{\hat{\mu} - \hat{r}_f}{\hat{\sigma}}$$

where

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{t=1}^n (R^t - \hat{\mu})^2$$

For ABC fund,  $S = 0.436$ . For S&P 500,  $S = 0.467$ . If you are a CAPM believer, you may think ABC fund is not efficient.

#### Testing CAPM Using Linear Regression

Suppose there are  $n$  risky assets in the market with return rates in the following period to be  $R_{i,t}, i = 1, \dots, n$ . Suppose the returns of the risk-free asset and of the market portfolio in the following period are  $r_f$  and  $R_M$ , respectively.

Decomposition:

$$R_i - r_f = \alpha_i + \beta_i (R_M - r_f) + \epsilon_i, \quad i = 1, \dots, n$$

where  $\epsilon_i$  are independent of each other and of  $R_M$ .

Vector form:

$$\mathbf{R} - r_f \mathbf{1}_n = \boldsymbol{\alpha} + \boldsymbol{\beta} (R_M - r_f) + \boldsymbol{\epsilon}$$

where  $\mathbf{R} := (R_1, \dots, R_n)^\top$ ,  $\boldsymbol{\alpha} := (\alpha_1, \dots, \alpha_n)^\top$ ,  $\boldsymbol{\beta} := (\beta_1, \dots, \beta_n)^\top$ , and  $\boldsymbol{\epsilon} := (\epsilon_1, \dots, \epsilon_n)^\top$ .  $\epsilon$  is independent of  $R_M$ .

CAPM holds if and only if  $\boldsymbol{\alpha} = 0$ . To test whether CAPM holds, suppose we have historical data of the asset returns:  $R_{i,t}, R_{j,t}, r_{f,t}, t = 1, \dots, T$ . Suppose the asset returns in different periods are i.i.d. Run vector linear regression to estimate  $\boldsymbol{\alpha}$  and  $\boldsymbol{\beta}$ . Reject CAPM if the estimate of  $\boldsymbol{\alpha}$  is significantly different from zero.

### 5.2 Arbitrage Pricing Theory

#### A Multi-Factor Model

Fix one period and consider  $n$  zero-cost strategies in this period with payoffs  $\tilde{r}_1, \dots, \tilde{r}_n$ , respectively. Assume the following factor model:

$$\tilde{r}_i = \alpha_i + \sum_{k=1}^m \beta_{ik} f_k + \epsilon_i, \quad i = 1, \dots, n$$

where  $s_k$  stands for the risk-free growth rate from the beginning of  $k$ -th period to the end, i.e.,  $s_k := \prod_{j=k}^{T-1} (1 + r_{f,j+1})$ , and  $\beta_k := \prod_{j=k}^{N-1} (1 - \mu_{j+1}^-) D_{j+1}^- \mu_{j+1}^-$ . Moreover, the optimal variance is  $2v_0(x_0)$  at the optimal  $\lambda_0$  and at the initial wealth  $x_0$ , i.e., the variance of the terminal wealth under the optimal strategy is:

$$Var(X_T) = \frac{\beta_1}{1 - \beta_0} (z_0 - s_0 x_0)^2$$

The above traces out the efficient frontier as  $z_0$  varies in  $[s_0 x_0, +\infty)$ . The mutual fund theorem still holds.

## 6.8 Continuous-Time Model

### Market

We can extend the multi-period model to a continuous-time model. Suppose the market has  $m$  risky stocks with prices at time  $t$  denoted as  $S_{i,t}$ ,  $i = 1, \dots, m$ , and a risk-free asset with price  $S_{0,t}$  at time  $t$ . Suppose that the risk-free asset generates net return rate  $rdt$  in an infinitesimally small time period  $[t, t + dt]$ , where  $r$  is a constant, i.e.,

$$dS_{0,t}/S_{0,t} = rdt$$

Suppose that each stock  $i$  generates excess return rate  $b_i dt + \sum_{j=1}^m \sigma_{i,j} dW_{j,t}$  for some constants  $b_i$  and  $\sigma_{i,j}$ ,  $j = 1, \dots, m$ , where  $(W_{1,t})_{t \geq 0}, \dots, (W_{m,t})_{t \geq 0}$  are  $m$  independent standard Brownian motions. In other words,

$$dS_{i,t}/S_{i,t} = (r + b_i)dt + \sum_{j=1}^m \sigma_{i,j} dW_{j,t}$$

### Portfolio

Assume that  $\sigma := (\sigma_{i,j})_{i,j=1,\dots,m}$  is invertible. Denote by  $\pi_t = (\pi_{1,t}, \dots, \pi_{m,t})^\top$  the agent's portfolio held in the infinitesimally small time period  $[t, t + dt]$  i.e.,  $\pi_{i,t}$  dollars are invested in stock  $i$ ,  $i = 1, \dots, m$ , and the remaining is invested in the risk-free asset in this period. Then, the agent's wealth, denoted as  $(X_t)_{t \geq 0}$ , evolves as:

$$dX_t = rX_t dt + \pi_t (bdt + \sigma dW_t)$$

where  $b := (b_1, \dots, b_m)^\top$  and  $W_t := (W_{1,t}, \dots, W_{m,t})^\top$ . Suppose the agent's wealth at the initial time 0 is  $x_0$ .

### Expected Utility Maximization

Consider maximization of expected utility of wealth at a given terminal time  $T$ :

$$\begin{aligned} \max_{\pi_t} & \mathbb{E}[u(X_T)] \\ \text{s.t.} \quad & dX_t = rX_t dt + \pi_t^\top (bdt + \sigma dW_t) \\ & X_0 = x_0 \end{aligned}$$

where  $u(\cdot)$  is the agent's utility function.

For power utility function  $u(x) = x^{1-\gamma}/(1-\gamma)$ ,  $x > 0$  for certain  $0 < \gamma \neq 1$ , the optimal portfolio is:

$$\pi_t^* = \frac{1}{\gamma} (\sigma \sigma^\top)^{-1} b, \quad t \in [0, T]$$

### Mean-Variance Analysis

Consider mean-variance portfolio selection. Suppose the target for the expected wealth at the terminal time  $T$  is  $z_0 > x_0 e^{rT}$ . The mean-variance problem is:

$$\begin{aligned} \max_{\pi_t} & (\pi_t)_{t \in [0, T]} \frac{1}{2} Var(X_T) \\ \text{s.t.} \quad & dX_t = rX_t dt + \pi_t^\top (bdt + \sigma dW_t) \\ & X_0 = x_0 \\ & \mathbb{E}[X_T] = z_0 \end{aligned}$$

The optimal portfolio is:

$$\pi_t^* = \left[ e^{-r(T-t)} z_0 - e^{-b^\top (\sigma \sigma^\top)^{-1} b T} e^{rT} x_0 - X_t \right] (\sigma \sigma^\top)^{-1} b, \quad t \in [0, T]$$

The optimal variance is:

$$Var(X_T) = \frac{e^{-b^\top (\sigma \sigma^\top)^{-1} b T}}{1 - e^{-b^\top (\sigma \sigma^\top)^{-1} b T}} (z_0 - e^{rT} x_0)^2$$

## 7 Limit Order Book

### 7.1 Bid, Ask, and Mid Prices

The lowest price among all sell limit orders is referred to as ask price.

The highest price among all buy limit orders is referred to as bid price.

The difference between the bid and ask prices is referred to as bid-ask spread. The average of the bid and ask prices is mid price.

### 7.2 Resolution Parameters

In general, an order  $x = (p_x, \omega_x, t_x)$  is submitted at time  $t_x$  with price  $p_x$  and size  $\omega_x > 0$  (respectively,  $\omega_x < 0$ ) is a commitment to sell (respectively, buy) up to  $|\omega_x|$  units of the traded at a price no less than (respectively, no greater than)  $p_x$ . The lot size  $\sigma$  of an LOB is the smallest amount of the asset that can be traded within it. All orders must arrive with a size  $\omega_k \in \{\pm k\sigma : k = 1, 2, \dots\}$ . The tick size  $\pi$  of an LOB is the smallest permissible price interval between different orders within it. All orders must arrive with a price that is specified to the accuracy of  $\pi$ .

The lot size  $\sigma$  and tick size  $\pi$  of an LOB are collectively called its resolution parameters.

### 7.3 Limit Order Book

An LOB  $\mathcal{B}(t)$  is the set of all active orders in a market at time  $t$ . The active orders in an LOB  $\mathcal{B}(t)$  can be partitioned into the set of active buy orders  $\mathcal{B}(t)$ , for which  $\omega_k < 0$ , and the set of active sell orders  $\mathcal{B}(t)$ , for which  $\omega_k > 0$ .

### 7.4 Depth

The bid-side depth available at price  $p$  and at time  $t$  is

$$n^b(p, t) := \sum_{\{x \in \mathcal{B}(t) : p_x = p\}} \omega_x$$

The ask-side depth available at price  $p$  and at time  $t$ , denoted  $n^a(p, t)$ , is defined similarly using  $\mathcal{a}(t)$ .

The bid price at time  $t$  is the highest stated price among all active buy orders at time  $t$ , i.e.,

$$b(t) := \max_{x \in \mathcal{B}(t)} p_x$$

The ask price at time  $t$  is the lowest stated price among all active sell orders at time  $t$ , i.e.,

$$a(t) := \min_{x \in \mathcal{a}(t)} p_x$$

### 7.5 Relative Price

The bid-ask spread at time  $t$  is  $s(t) := a(t) - b(t)$ . The mid price at time  $t$  is  $m(t) := [a(t) + b(t)]/2$ .

For a given price  $p$ , the bid-relative price is  $\delta^b(p) := b(t) - p$  and the ask-relative price is  $\delta^a(p) := p - a(t)$ .

For a given order  $x = (p_x, \omega_x, t_x)$ , the relative price of the order is  $\delta_x := \delta^b(p_x)$  if the order is a buy order, and  $\delta_x := \delta^a(p_x)$  if the order is a sell order.

The bid-side depth available at relative price  $\delta$  and at time  $t$  is

$$N^b(\delta, t) = \sum_{\{x \in \mathcal{B}(t) : \delta^b = \delta\}} \omega_x$$

The ask-side is defined similarly.

### 7.6 Price changes in LOBs

Consider a buy (respectively, sell) order  $x = (p_x, \omega_x, t_x)$  that arrives immediately after time  $t$ . If  $p_x \leq b(t)$  (respectively,  $p_x \geq a(t)$ ), then  $x$  is a limit order that becomes active upon arrival. It does not cause  $b(t)$  or  $a(t)$  to change.

If  $b(t) < p_x < a(t)$ , then  $x$  is a limit order that becomes active upon arrival. Upon arrival,  $b(t_x) = p_x$  (respectively,  $a(t_x) = p_x$ ).

If  $p_x \geq a(t)$  (respectively,  $p_x \leq b(t)$ ), then  $x$  is a market order that immediately matches to one or more active sell (respectively, buy) orders upon arrival. Whenever such a matching occurs, it does so at the price of the active order, which is not necessarily equal to the price of the incoming order. Whether or not such a matching causes  $a(t)$  (respectively,  $b(t)$ ) to change at time  $t_x$  depends on  $n^a(a(t), t)$  (respectively,  $n^b(b(t), t)$ ) and  $\omega_x$ .

In particular, the new bid price  $b(t_x)$  upon arrival of a sell market order  $x$  is

$$b(t_x) = \max(p_x, q), \text{ where } q = \arg\max_{k'} n^b(k', t) |n^b(k', t)| > |\omega_x|$$

Similarly, the new ask price  $a(t_x)$  upon arrival of a buy market order  $x$  is

$$a(t_x) = \min(p_x, q), \text{ where } q = \arg\min_{k'} \sum_{k=a(t)}^k |n^a(k', t)| > |\omega_x|$$

Put another way, the incoming order  $x$  matches to the highest priority active order  $y$  of opposite type.

## 8 Sample Final Exam Paper

### Question 1

Consider P&L  $Z$  with the following distribution:

$$\begin{aligned} \mathbb{P}(Z = -12) &= 0.01, \mathbb{P}(Z = -6) = 0.02, \mathbb{P}(Z = -4) = 0.01, \mathbb{P}(Z = -1) = 0.02 \\ \mathbb{P}(Z = 0) &= 0.5, \mathbb{P}(Z = 1) = 0.24, \mathbb{P}(Z = 3) = 0.2 \end{aligned}$$

Compute the 95%-VaR, the 95% CVaR, the expected loss conditional on the loss being larger than or equal to the 95%-VaR, and the expected loss conditional on the loss being strictly larger than the 95%-VaR.

### Solution 1

We compute the CDF of  $Z$ , denoted as  $F_{-Z}(x)$ . Calculation yields:

$$F_{-Z}(x) = \begin{cases} 0, & x < -3, \\ 0.2, & x \in [-3, -1), \\ 0.44, & x \in [-1, 0), \\ 0.94, & x \in [0, 1), \\ 0.96, & x \in [1, 4), \\ 0.97, & x \in [4, 6), \\ 0.99, & x \in [6, 12), \\ 1, & x \geq 12. \end{cases}$$

As a result, we have:

$$VaR_\alpha(Z) = \inf\{x \in \mathbb{R} : \mathbb{P}(-Z \leq x) \geq \alpha\} = \inf\{x \in \mathbb{R} : F_{-Z}(x) \geq \alpha\}$$

Specifically for different  $\alpha$ :

$$VaR_\alpha(Z) = \begin{cases} -3, & \alpha \in [0, 0.2], \\ -1, & \alpha \in (0.2, 0.44], \\ 0, & \alpha \in (0.44, 0.94], \\ 1, & \alpha \in (0.94, 0.96], \\ 4, & \alpha \in (0.96, 0.97], \\ 6, & \alpha \in (0.97, 0.99], \\ 12, & \alpha \in (0.99, 1). \end{cases}$$

Consequently,  $VaR_{0.95\%}(Z) = 1$ .

The 95% CVaR is calculated as:

$$\begin{aligned} CVaR_{0.95\%}(Z) &= \frac{1}{1 - 0.95} \int_{0.95}^1 VaR_\alpha(Z) d\beta \\ &= \frac{1}{0.05} \times [1 \times 0.01 + 4 \times 0.01 + 6 \times 0.02 + 12 \times 0.01] = 5.8 \end{aligned}$$

The expected loss conditional on the loss being larger than or equal to the 95%-VaR ( $VaR_{0.95\%} = 1$ ) is:

$$\begin{aligned} \mathbb{E}[-Z | -Z \geq VaR_{0.95\%}(Z)] &= \mathbb{E}[-Z | -Z \geq 1] \\ &= \frac{1}{0.02 + 0.01 + 0.02 + 0.01} \times [1 \times 0.02 + 4 \times 0.01 + 6 \times 0.02 + 12 \times 0.01] = 5 \end{aligned}$$

The expected loss conditional on the loss being strictly larger than the 95%-VaR is:

$$\begin{aligned} \mathbb{E}[-Z | -Z > VaR_{0.95\%}(Z)] &= \mathbb{E}[-Z | -Z > 1] \\ &= \frac{1}{0.01 + 0.02 + 0.01} \times [4 \times 0.01 + 6 \times 0.02 + 12 \times 0.01] = 7 \end{aligned}$$

### Question 2

Consider a wealth manager who is looking into a particular portfolio strategy. She found that the historical estimates of the mean and variance of the portfolio's monthly return are 0.7% and 1.8%, respectively. She also estimated the mean and variance of the market portfolio's monthly return to be 0.6% and 1.2%, respectively, and the correlation between the return rates of the portfolio and the market is 0.9. The monthly risk-free return rate is flat at 0.2% in the past.

(a) According to CAPM, is this strategy a good one? Why?

(b) The director of the wealth manager believes that the Fama-French three-factor model instead of CAPM should be used to assess the portfolio strategy. To this end, the wealth manager estimated the loadings of the portfolio for SMB and HML, which turn out to be -0.3 and 0.7, respectively. She also estimated the expected payoffs for the zero-cost strategies associated with SMB and HML to be 0.4% and 0.3%, respectively. Is this strategy a good one under the Fama-French three-factor model? Why?

### Solution 2

(a) We compute  $\beta_1$  first:

$$\beta_1 = \frac{\text{cov}(r_p, r_M)}{\text{Var}(r_M)} = \frac{\rho \sigma_p \sigma_M}{\sigma_M^2} = \frac{0.9 \times \sqrt{1.8\%} \times \sqrt{1.2\%}}{1.2\%} = 1.10227$$

As a result, the risk premium of the portfolio strategy implied by CAPM is:

$$\bar{r}_1 - r_f = \beta_1 (r_M - r_f) = 1.10227 \times (0.6\% - 0.2\%) = 0.44091\%$$

This is lower than the estimated risk premium, namely  $0.7\% - 0.2\% = 0.5\%$ . So under CAPM, this portfolio strategy is good (it offers a positive alpha).

(b) The loadings of the strategy on the market risk factor, SMB, and HML are 1.10227, -0.3, and 0.7, respectively. The risk premiums of these three factors are 0.6%-0.2%, 0.4%, and 0.3%, respectively. Thus, the model risk premium of the strategy under the Fama-French three-factor model is:

$$1.10227 \times (0.6\% - 0.2\%) - 0.3 \times 0.4\% + 0.7 \times 0.3\% = 0.531\%$$

This is higher than the estimated risk premium of 0.5%. Thus, this strategy is a bad one when benchmarked to the Fama-French three-factor model.

### Question 3

Suppose that David's total wealth is \$40,000, in which his car is worth \$15,000. Suppose his utility function is  $u(x) = 2\sqrt{x}$ . There is a 5% probability that his car would be stolen and he will lose \$15,000. There is also a 10% probability that his car would be damaged in an accident and he will have to spend \$10,000 to repair it.

(a) What is the Arrow-Pratt absolute risk aversion coefficient of David?

(b) If he wants to buy a car insurance which can fully cover his loss in both cases, i.e., stolen and damaged, what is the maximum amount of money he is willing to pay for the insurance?

### Solution 3

(a) The Arrow-Pratt absolute risk aversion coefficient is:

$$r(x) = -\frac{u''(x)}{u'(x)} = -\frac{2 \times (1/2) \times (-1/2) x^{-3/2}}{2 \times (1/2) x^{-1/2}} = \frac{1}{x}$$

(b) The maximum amount of money,  $A$ , David is willing to pay is the amount such that David is indifferent between his wealth if he purchases the insurance contract and his wealth if he does not purchase the contract.

If David purchases the insurance contract, his wealth is  $40000 - A$  and the utility of the wealth is  $2\sqrt{40000 - A}$ .

If David does not purchase the insurance contract, the expected utility of his wealth is:

$$2\sqrt{40000 - 15000 \times 0.05 + 2\sqrt{40000 - 10000 \times 0.10} + 2\sqrt{40000 \times 0.85} = 390.4524$$

Setting the expected utility of the wealth in the case in which David purchases the insurance equal to the case in which he does not purchase the insurance, namely:

$$2\sqrt{40000 - A} = 390.4524$$

We derive  $A = 1886.731$ .

### Question 4

Consider the following Limit Order Book (LOB) for a security:

**Bid Side (Depth Available at Price):** \$1.99 (Depth: 2), \$1.98 (Depth: 3), \$1.97 (Depth: 1), \$1.96 (Depth: 2), \$1.95 (Depth: 4).

**Ask Side (Depth Available at Price):** \$2.02 (Depth: 1), \$2.03 (Depth: 2), \$2.04 (Depth: 2), \$2.05 (Depth: 3), \$2.06 (Depth: 4).

Current Bid Price: \$1.99. Current Ask Price: \$2.02. Current Mid Price: \$2.005.

(a) Suppose an order arrives to buy 3 units of the security at price \$2.00. Calculate the mid price immediately after the arrival of the order.

(b) Suppose an order arrives to sell 1 unit of the security at price \$2.02. Calculate the mid price immediately after the arrival of the order.

(c) Suppose an order arrives to buy 3 units of the security at the market price. Calculate the mid price immediately after the arrival of the order.

(d) Suppose cancellation of an existing sell orders of size 1 at price \$2.02 arrives. Calculate the mid price immediately after the arrival of cancellation.

### Solution 4

(a) The order is a buy limit order at \$2.00. Since \$2.00 < Ask (\$2.02), it becomes the new best bid. New Bid: \$2.00. Ask remains: \$2.02. Mid Price:

$$\frac{2.02 + 2.00}{2} = \$2.01$$

(b) The order is a sell limit order at \$2.02. Since \$2.02 > Bid (\$1.99) and \$2.02 is the current Ask, it adds depth to the ask side but does not change the best prices. Bid: \$1.99. Ask: \$2.02. Mid Price remains:

$$\frac{2.02 + 1.99}{2} = \$2.005$$

(c) A market buy order for 3 units will walk up the book. It consumes the 1 unit at \$2.02. It consumes 2 units at \$2.03. The new best Ask becomes \$2.04. Bid remains \$1.99. Mid Price:

$$\frac{2.04 + 1.99}{2} = \$2.015$$

(d) Cancellation of 1 unit at \$2.02. The existing depth at \$2.02 was 1. After cancellation, depth at \$2.02 becomes 0. The next available ask is \$2.03. New Ask: \$2.03. Bid remains \$1.99. Mid Price:

$$\frac{2.03 + 1.99}{2} = \$2.01$$

### Question 5

Consider a 3-period mean-variance portfolio selection problem with one risk-free asset and 1 risky stock. Suppose the initial wealth is \$1. Suppose that the risk-free net return in each period is  $r_{f,k} = 0.03$ . In addition, for  $k = 1, \dots, 3$  the expected excess return vector of the stocks is:

$$\mu_k = \mathbb{E}[R_k] = 5\%$$

and variance of the stock excess return rate is:

$$C_k = \text{Var}(R_k) = 0.09$$

Suppose the agent's target is to grow the wealth by 6% in each period. Find the optimal portfolio strategy of the agent.

### Solution 5

The number of investment periods is  $N = 3$ . We first compute  $D_k$ :

$$D_k = \mathbb{E}[R_k R_k^\top] = C_k + \mu_k \mu_k^\top = (5\%)^2 + 0.09 = 0.0925$$

$$\mu_k^\top D_k^{-1} \mu_k = (5\%)^2 / 0.0925 = 0.027027028, \quad k = 1, \dots, 3$$

Next, we compute:

$$D_k^{-1} \mu_k = 5\% / 0.0925 = 0.54054054$$

$$s_k = \prod_{j=k}^{N-1} (1 + r_{f,j+1}) = 1.03^{3-k}, \quad k = 0, \dots, 2$$

$$\beta_k = \prod_{j=k}^{N-1} (1 - \mu_{j+1}^-) D_{j+1}^- \mu_{j+1}^- = 0.972972973^{3-k}, \quad k = 0, \dots, 2$$

The target expected final wealth is  $z_0 = 1 \times (1 + 0.06)^3$ . (The problem says "grow the wealth by 6% in each period", implying  $z_0 = (1.06)^3$ ).

The optimal strategy formula is  $\pi_k^* = -A_k x_k + B_k$ .

We calculate the coefficients:

$$A_k := (1 + r_{f,k+1}) D_{k+1}^- \mu_{k+1}^- = 1.03 \times 0.54054054 = 0.556756757, \quad k = 0, 1, 2$$

$$B_k := A_k s_k^{-1} \frac{z_0 - \beta_0 x_0}{1 - \beta_0}$$

Substituting the values ( $x_0 = 1$ ,  $z_0 = (1.06)^3$ ,  $s_0 = 1.03^3$ ,  $\beta_0 = 0.972973$ ):

$$\begin{aligned} B_k &= 0.556756757 \times \frac{(1.06)^3 - 0.972972973 \times 1.03^3 \times 1}{1 - 0.972972973} \times 1.03^{3-k} \\ &= 1.30187509 \times 1.03^{3-k} \end{aligned}$$

Thus, the values for  $B_k$  are:

$$B_k \approx \begin{cases} 1.91400139, & k = 0, \\ 1.227142143, & k = 1, \\ 1.263956407, & k = 2. \end{cases}$$